

角动量链接网络宇宙论：ANG 公理体系与物理本体论

Angular Momentum Link Network Cosmology: The ANG Axiomatic System and Physical Ontology

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摘要 Abstract

中文

本文提出宇宙的本体论模型：宇宙是由角动量链接杆组成的网络，最底层节点质量为零，链接杆包含四个基本属性——能量（角动量）、长度、角度、相位。一个局域空间闭合体，其内部链接网络构成质量，链接在闭合体界面的分布构成相互作用力，闭合体的运动构成能量，而时间是相位的演化速率。基于此，我们建立了六条公理（A1-A6）构成完整的 ANG 公理体系。该系统统一了关系、距离、刚度、方向与演化，并自然退化到广义相对论、量子场论和经典力学。

English

This paper proposes an ontological model of the universe: the universe is a network of angular momentum link rods, with zero-mass nodes at the lowest level. Each link rod possesses four fundamental properties—energy (angular momentum), length, angle, and phase. A localized closed spatial manifold has its internal link network constituting mass, the distribution of links on the closed manifold's boundary constituting interaction forces, the motion of the closed manifold constituting energy, and time being the rate of phase evolution. Based on this, we establish six axioms (A1–A6) forming the complete ANG (Angular Momentum Link Network Geometry) axiomatic system. This system unifies relation, distance, stiffness, direction, and evolution, and naturally degenerates to general relativity, quantum field theory, and classical mechanics.

1 引言：宇宙作为角动量链接网络

Introduction: The Universe as an Angular Momentum Link Network

1.1 核心本体论命题

1.1 Core Ontological Proposition

中文

我们提出以下五个基本命题，构成 ANG 宇宙论的物理基础：

命题	陈述	对应物理对象
P1	宇宙底层由角动量链接杆组成的网络构成	场的基态结构
P2	最底层节点质量为零	真空/无质量激发
P3	链接杆包含四个属性：能量、长度、角度、相位	相互作用的基本单元
P4	局域空间闭合体内部的链接网络构成质量	物质/能量密度
P5	链接在闭合体界面的分布构成相互作用力	力/场

English

We put forward the following five fundamental propositions, which form the physical foundation of ANG cosmology:

Proposition	Statement	Corresponding Physical Object
P1	The fundamental layer of the universe consists of a network built from	Ground state structure of fields

	angular momentum link rods	
P2	Nodes at the lowest level carry zero mass	Vacuum / massless excitation
P3	Each link rod has four attributes: energy, length, angle, phase	Elementary unit of interactions
P4	The internal link network of a localized closed spatial manifold forms mass	Matter / energy density
P5	The distribution of links on the boundary of closed manifolds generates interaction forces	Force / physical field

1.2 物理量到 ANG 元素的统一映射

1.2 Unified Mapping from Physical Quantities to ANG Elements

中文

物理量	ANG 映射	对应公理
质量	闭合链接网络的总能量	Axiom A2
能量	链接的角动量含量 $\mathcal{N} = \sqrt{L^2 + S^2}$	Axiom A1/A2
力	界面链接的应力能分布 $s_{ij} = \kappa_{ij} \ell_{ij}$	Axiom A5
运动	链接重连序列 $\mathcal{R}_k$	Axiom A5

时间	相位演化速率 $\omega = d\phi/dt$	Axiom A5 (时间标度)
长度	链接几何距离 ℓ_{ij}	Axiom A3
角度	链接方向夹角 θ_{ij}	Axiom A4
相位	链接方向角的累积 $\phi_{ij} = \int \omega dt$	Axiom A4/A5

English

Physical Quantity	ANG Mapping	Corresponding Axiom
Mass	Total energy of closed link networks	Axiom A2
Energy	Angular momentum content of links $\mathcal{N} = \sqrt{L^2 + S^2}$	Axiom A1/A2
Force	Stress-energy distribution of boundary links $s_{ij} = \kappa_{ij}\ell_{ij}$	Axiom A5
Motion	Link reconnection sequence $\mathcal{R}_k$	Axiom A5
Time	Phase evolution rate $\omega = d\phi/dt$	Axiom A5 (time scaling)
Length	Geometric distance of link ℓ_{ij}	Axiom A3
Angle	Directional angle between links θ_{ij}	Axiom A4
Phase	Accumulation of directional angles $\phi_{ij} = \int \omega dt$	Axiom A4/A5

2 ANG 公理体系

2 The ANG Axiomatic System

Axiom A1 (网络本体公理)

Axiom A1 (Network Ontology Axiom)

中文

宇宙是一个角动量链接网络，由节点集 $\mathcal{V}$ 、链接集 $\mathcal{E}$ 、方向角场 Θ 、重连算子序列 $\mathcal{R}$ 构成：

$$\mathcal{G}_{\text{Universe}} = \{\mathcal{V}, \mathcal{E}, \Theta, \mathcal{R}\}$$

- 节点 $\mathcal{N}_i$ ：最底层质量为零
- 链接 $\mathcal{E}_{ij}$ ：角动量链接杆

$$\mathcal{E}_{ij} \equiv \{\ell_{ij}, \kappa_{ij}, \theta_{ij}, \phi_{ij}\}$$

其中：

ℓ_{ij} ：长度（几何距离）

κ_{ij} ：刚度（强度/耦合系数）

θ_{ij} ：方向角（空间取向）

ϕ_{ij} ：相位（动力学状态）

English

The universe forms an angular momentum link network consisting of node set $\mathcal{V}$, link set $\mathcal{E}$, directional angle field Θ , and sequence of reconnection operators $\mathcal{R}$:

$$\mathcal{G}_{\text{Universe}} = \{\mathcal{V}, \mathcal{E}, \Theta, \mathcal{R}\}$$

- Node $\mathcal{N}_i$: zero mass at the fundamental level
- Link $\mathcal{E}_{ij}$: angular momentum link rod

$$\mathcal{E}_{ij} \equiv \{\ell_{ij}, \kappa_{ij}, \theta_{ij}, \phi_{ij}\}$$

Definitions:

ℓ_{ij} : length (geometric distance)

κ_{ij} : stiffness (strength / coupling coefficient)

θ_{ij} : directional angle (spatial orientation)

ϕ_{ij} : phase (dynamical state)

Axiom A2 (角动量守恒公理)

Axiom A2 (Angular Momentum Conservation Axiom)

中文

节点和链接的总角动量严格守恒：

$$\sum_{i=1}^N \mathcal{N}_i + \sum_{i,j} \oint_{\partial v} \varepsilon_{ij} \cdot d\mathbf{S} = 0$$

物理含义：

- 第一项：所有节点的内禀角动量之和
- 第二项：所有链接交换的角动量通量之和
- 最底层节点 $\mathcal{N}_i \rightarrow 0$ ，所有物理内容由链接携带

English

The total angular momentum of nodes and links is strictly conserved:

$$\sum_{i=1}^N \mathcal{N}_i + \sum_{i,j} \oint_{\partial v} \varepsilon_{ij} \cdot d\mathbf{S} = 0$$

Physical Interpretation:

- Term 1: Sum of intrinsic angular momentum of all nodes
- Term 2: Integral flux of angular momentum exchanged via all links
- Fundamental nodes satisfy $\mathcal{N}_i \rightarrow 0$; all physical information is carried by links.

Axiom A3 (度量几何公理)

Axiom A3 (Metric Geometry Axiom)

中文

空间几何由链接的长度和方向决定，满足广义勾股关系：

$$\ell_{ij}^2 + \ell_{jk}^2 - 2\ell_{ij}\ell_{jk}\cos\theta_{ijk} = \ell_{ik}^2$$

连续极限：

$$g_{\mu\nu}(x) = \lim_{\text{mesh} \rightarrow 0} \left(\frac{1}{\kappa} \frac{\partial^2 \mathcal{E}}{\partial x^\mu \partial x^\nu} \right)$$

物理含义：空间的弯曲由链接网络的分布决定——这正是广义相对论的离散形式。

English

Spatial geometry is determined by link lengths and orientations, obeying the generalized Pythagorean relation:

$$\ell_{ij}^2 + \ell_{jk}^2 - 2\ell_{ij}\ell_{jk}\cos\theta_{ijk} = \ell_{ik}^2$$

Continuum limit:

$$g_{\mu\nu}(x) = \lim_{\text{mesh} \rightarrow 0} \left(\frac{1}{\bar{\kappa}} \frac{\partial^2 \mathcal{E}}{\partial x^\mu \partial x^\nu} \right)$$

Physical Interpretation: Spacetime curvature arises from the distribution of link networks; this expression constitutes the discrete counterpart of general relativity.

Axiom A4 (方向性与相位公理)

Axiom A4 (Directionality and Phase Axiom)

中文

方向角 θ_{ij} 与相位 ϕ_{ij} 由链接刚度的梯度决定：

$$\theta_{ij} = \arccos \left(\frac{\mathbf{n}_{ij} \cdot \mathbf{n}_{ik}}{\|\mathbf{n}_{ij}\| \|\mathbf{n}_{ik}\|} \right)$$

$$\phi_{ij}(t) = \phi_{ij}(0) + \int_0^t \omega_{ij}(t') dt'$$

物理含义：

- 方向角决定空间取向（力方向）
- 相位累积决定演化状态（时间）
- 相位的演化速率即为时间

English

Directional angle θ_{ij} and phase ϕ_{ij} are governed by the gradient of link stiffness:

$$\theta_{ij} = \arccos \left(\frac{\mathbf{n}_{ij} \cdot \mathbf{n}_{ik}}{\|\mathbf{n}_{ij}\| \|\mathbf{n}_{ik}\|} \right)$$

$$\phi_{ij}(t) = \phi_{ij}(0) + \int_0^t \omega_{ij}(t') dt'$$

Physical Interpretation:

- Directional angles determine spatial orientation (force direction)
- Cumulative phase encodes evolutionary state (time)
- Time is defined as the evolution rate of phase.

Axiom A5 (重连动力学公理)

Axiom A5 (Reconnection Dynamics Axiom)

中文

闭合体的运动、能量、时间由链接的重连驱动：

$$s_{ij} = \kappa_{ij} \cdot \ell_{ij} > \tau_{\max} \implies \mathcal{R}_k: \mathcal{E}_{ij} \rightarrow \mathcal{E}_{ik}$$

$$P_{\text{reconn}}(j \rightarrow k) = \frac{\exp(-\beta \cdot \ell_{ik})}{\sum_m \exp(-\beta \cdot \ell_{im})}$$

$$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

物理含义：

- 链接断裂与重连 = 物理系统的演化
- 重连频率 = 能量转换速率
- 时间 = 相位演化速率

English

Motion, energy and time of closed manifolds are driven by link reconnection:

$$S_{ij} = \kappa_{ij} \cdot \ell_{ij} > \tau_{\max} \Rightarrow \mathcal{R}_k: \mathcal{E}_{ij} \rightarrow \mathcal{E}_{ik}$$

$$P_{\text{reconn}}(j \rightarrow k) = \frac{\exp(-\beta \cdot \ell_{ik})}{\sum_m \exp(-\beta \cdot \ell_{im})}$$

$$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$$

Physical Interpretation:

- Link breaking and reconnection = evolution of physical systems
- Reconnection frequency = energy conversion rate
- Time = phase evolution rate.

Axiom A6（层级闭合公理）

Axiom A6 (Hierarchical Closure Axiom)

中文

闭合体的质量由内部链接网络的拓扑不变量决定：

$$M_{\text{body}} = \frac{1}{c^2} \sum_{\text{内部链接}} \mathcal{N}_{ij}$$

$$\mathbf{F}_{\text{surface}} = \oint_{\partial \mathcal{V}} \mathcal{E}_{ij} \cdot d\mathbf{S}$$

$$E_{\text{motion}} = \sum_{\text{重连事件}} \Delta \mathcal{N}_{ij}$$

物理含义：

- 闭合体内部链接总角动量 → 质量
- 闭合体界面上的链接分布 → 力

- 重连事件的总角动量变化 → 能量
- 相位的演化速率 → 时间

English

The mass of a closed manifold is determined by topological invariants of its internal link network:

$$M_{\text{body}} = \frac{1}{c^2} \sum_{\text{internal links}} \mathcal{N}_{ij}$$

$$\mathbf{F}_{\text{surface}} = \oint_{\partial\mathcal{V}} \boldsymbol{\varepsilon}_{ij} \cdot d\mathbf{S}$$

$$E_{\text{motion}} = \sum_{\text{reconnection events}} \Delta \mathcal{N}_{ij}$$

Physical Interpretation:

- Total angular momentum of internal links within closed manifold → **Mass**
- Distribution of links on manifold boundary → **Force**
- Total angular momentum variation from reconnection events → **Energy**
- Phase evolution rate → **Time**

3 五命题的数学证明

3 Proof of the Five Propositions

定理 1：质量 = 闭合链接网络（命题 P4）

Theorem 1: Mass = Closed Link Network (Proposition P4)

中文

陈述：闭合体 $\mathcal{V}$ 的质量由内部链接网络的总能量决定。

证明：

由 Axiom A6，闭合体内部质量：

$$M_{\mathcal{V}} = \frac{1}{c^2} \sum_{\boldsymbol{\varepsilon}_{ij} \in \mathcal{V}} \mathcal{N}_{ij}$$

其中 $\mathcal{N}_{ij} = \sqrt{L_{ij}^2 + S_{ij}^2}$ 。

当 $\mathcal{V}$ 为空（无内部链接）时：

$$M_{\mathcal{V}} = 0$$

满足 P2。当 $\mathcal{V}$ 包含闭合链接网络时，质量大于零。

English

Statement: The mass of closed manifold $\mathcal{V}$ is determined by the total energy of its internal link network.

Proof:

From Axiom A6, the mass of closed manifold:

$$M_{\mathcal{V}} = \frac{1}{c^2} \sum_{\varepsilon_{ij} \subset \mathcal{V}} \mathcal{N}_{ij}$$

$$\text{where } \mathcal{N}_{ij} = \sqrt{L_{ij}^2 + S_{ij}^2}.$$

If $\mathcal{V}$ is empty (no internal links):

$$M_{\mathcal{V}} = 0$$

Consistent with P2. If $\mathcal{V}$ contains closed link networks, mass is positive.

定理 2：力 = 界面链接分布（命题 P5）

Theorem 2: Force = Boundary Link Distribution (Proposition P5)

中文

陈述：闭合体表面的力由界面链接的应力能决定。

证明：

由 Axiom A5，每根界面链接的应力能：

$$s_{ij}^{\text{boundary}} = \kappa_{ij} \cdot \ell_{ij}$$

总应力：

$$\mathbf{F}_{\text{surface}} = \oint_{\partial \mathcal{V}} \mathbf{n}_{ij} \cdot s_{ij}^{\text{boundary}} dS$$

方向由 θ_{ij} 决定。这正是场论中场的边界通量在离散网络上的直接对应——力是闭合体边界上链接应力能密度的定向积分。

English

Statement: The force acting on the surface of a closed manifold is determined by stress energy carried by boundary links.

Proof:

From Axiom A5, stress energy of each boundary link:

$$s_{ij}^{\text{boundary}} = \kappa_{ij} \cdot \ell_{ij}$$

Total stress:

$$\mathbf{F}_{\text{surface}} = \oint_{\partial V} \mathbf{n}_{ij} \cdot s_{ij}^{\text{boundary}} dS$$

Direction is governed by θ_{ij} . This expression is the discrete network counterpart of boundary flux in field theory: force equals the oriented integral of link stress-energy density over the manifold boundary.

定理 3：能量 = 运动 = 重连事件（命题 P4/P5 联合）

Theorem 3: Energy = Motion = Reconnection Events (Joint Proposition P4/P5)

中文

陈述：闭合体的运动能量由重连事件的总角动量变化决定。

证明：

由 Axiom A5，每次重连事件改变链接的角动量：

$$\Delta E_{\text{reconn}} = \Delta \mathcal{N}_{ij}$$

总能量：

$$E_{\text{motion}} = \sum_k \Delta \mathcal{N}_{ij}^{(k)}$$

其中 k 遍历所有重连事件。若无重连（静态闭合体）， $E_{\text{motion}} = 0$ 。

English

Statement: Kinetic energy of closed manifolds is determined by total angular momentum variation from reconnection events.

Proof:

From Axiom A5, each reconnection event modifies link angular momentum:

$$\Delta E_{\text{reconn}} = \Delta \mathcal{N}_{ij}$$

Total energy:

$$E_{\text{motion}} = \sum_k \Delta \mathcal{N}_{ij}^{(k)}$$

where k runs over all reconnection events. If no reconnection occurs (static closed manifold), $E_{\text{motion}} = 0$.

定理 4：时间 = 相位演化速率（命题 P4/P5 联合）

Theorem 4: Time = Phase Evolution Rate (Joint Proposition P4/P5)

中文

陈述：时间是链接相位的演化速率。

证明：

由 Axiom A4, 相位演化：

$$\phi_{ij}(t) = \phi_{ij}(0) + \int_0^t \omega_{ij}(t') dt'$$

由 Axiom A5:

$$\omega_{ij}(t) = \frac{1}{\Delta t} \cdot \Delta \theta_{ij}$$

全网平均时间：

$$t = \frac{1}{N_{\varepsilon}} \sum_{ij} \int \frac{d\phi_{ij}}{\omega_{ij}}$$

物理含义：当所有链接的相位停止演化 ($\omega = 0$)，时间停止。这准确地描述了“时间来源于运动（相位变化）”的物理直觉。

English

Statement: Time is defined as the evolution rate of link phase.

Proof:

From Axiom A4, phase evolution relation:

$$\phi_{ij}(t) = \phi_{ij}(0) + \int_0^t \omega_{ij}(t') dt'$$

From Axiom A5:

$$\omega_{ij}(t) = \frac{1}{\Delta t} \cdot \Delta \theta_{ij}$$

Globally averaged time:

$$t = \frac{1}{N_{\varepsilon}} \sum_{ij} \int \frac{d\phi_{ij}}{\omega_{ij}}$$

Physical Interpretation: Time freezes if phase evolution of all links halts ($\omega = 0$). This rigorously formalizes the physical intuition that time emerges from motion (phase variation).

4 与现有物理理论的统一

4 Unification with Established Physical Theories

4.1 广义相对论 ($N \rightarrow \infty$ 极限)

4.1 General Relativity ($N \rightarrow \infty$ Continuum Limit)

中文

当节点数 $N \rightarrow \infty$ 时：

$$g_{\mu\nu}(x) = \lim_{\text{mesh} \rightarrow 0} \left(\frac{1}{\kappa} \frac{\partial^2 \mathcal{E}}{\partial x^\mu \partial x^\nu} \right)$$

爱因斯坦场方程作为 Axiom A3 + A5 的联合推论：

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

English

As the number of nodes $N \rightarrow \infty$:

$$g_{\mu\nu}(x) = \lim_{\text{mesh} \rightarrow 0} \left(\frac{1}{\kappa} \frac{\partial^2 \mathcal{E}}{\partial x^\mu \partial x^\nu} \right)$$

The Einstein field equations emerge as a joint corollary of Axiom A3 and A5:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

4.2 量子场论（链接激发）

4.2 Quantum Field Theory (Link Excitations)

中文

链接的激发态 $\Delta\mathcal{E}_{ij}$ 对应量子涨落：

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

由 Axiom A5 的时间标度自然给出。

English

Excited link states $\Delta\mathcal{E}_{ij}$ correspond to quantum fluctuations:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

This uncertainty relation follows naturally from the time scaling defined in Axiom A5.

4.3 量子引力（网络重连）

4.3 Quantum Gravity (Network Reconnection)

中文

重连事件 $\mathcal{R}_k$ 是量子引力中的自旋网络演化：

$$\mathcal{R}_k: \mathcal{E}_{ij} \rightarrow \mathcal{E}_{ik} + \mathcal{E}_{jl}$$

此重连算子是圈量子引力中自旋网络顶点演化的 ANG 表述。

English

Reconnection event $\mathcal{R}_k$ describes spin network evolution in quantum gravity:

$$\mathcal{R}_k: \mathcal{E}_{ij} \rightarrow \mathcal{E}_{ik} + \mathcal{E}_{jl}$$

This reconnection operator provides an ANG formulation for vertex evolution within loop quantum gravity.

4.4 热力学第二定律（重连不可逆性）

4.4 Second Law of Thermodynamics (Irreversibility of Reconnection)

中文

重连概率分布:

$$P_{\text{reconn}}(j \rightarrow k) = \frac{e^{-\beta \ell_{ik}}}{\sum e^{-\beta \ell_{im}}}$$

重连向高熵状态演化, 熵增与时间方向一致:

$$\frac{dS}{dt} = k_B \ln \left(\frac{\text{重连后状态数}}{\text{重连前状态数}} \right) \geq 0$$

English

Reconnection probability distribution:

$$P_{\text{reconn}}(j \rightarrow k) = \frac{e^{-\beta \ell_{ik}}}{\sum e^{-\beta \ell_{im}}}$$

Reconnection drives the system toward higher-entropy configurations, and entropy growth aligns with the arrow of time:

$$\frac{dS}{dt} = k_B \ln \left(\frac{\text{Number of states after reconnection}}{\text{Number of states before reconnection}} \right) \geq 0$$

5 物理量-ANG 元素完整映射表

5 Complete Mapping Table: Physical Concepts and ANG Elements

中文

物理概念	ANG 元素	公理来源
宇宙本体	$\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \Theta, \mathcal{R}\}$	Axiom A1
质量 (闭合体)	$M = \frac{1}{c^2} \sum_{\mathcal{E} \in \mathcal{V}} \mathcal{N}$	Axiom A6
力 (界面)	$\mathbf{F} = \oint_{\partial \mathcal{V}} \boldsymbol{\varepsilon} \cdot d\mathbf{S}$	Axiom A6
能量 (运动)	$E = \sum \Delta \mathcal{N}_{ij}$	Axiom A5

时间（相位演化）	$t = \int d\phi/\omega$	Axiom A4/A5
长度	ℓ_{ij}	Axiom A3
方向	θ_{ij}	Axiom A4
相位	ϕ_{ij}	Axiom A4
相互作用强度	κ_{ij}	Axiom A5
应力	$s_{ij} = \kappa_{ij}\ell_{ij}$	Axiom A5
拓扑不变量	β_n, χ	Axiom A6

English

Physical Concept	ANG Element	Source Axiom
Universe Ontology	$\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \Theta, \mathcal{R}\}$	Axiom A1
Mass (Closed Manifold)	$M = \frac{1}{c^2} \sum_{\mathcal{E} \in \mathcal{V}} \mathcal{N}$	Axiom A6
Force (Boundary)	$\mathbf{F} = \oint_{\partial \mathcal{V}} \boldsymbol{\varepsilon} \cdot d\mathbf{S}$	Axiom A6
Energy (Motion)	$E = \sum \Delta \mathcal{N}_{ij}$	Axiom A5
Time (Phase Evolution)	$t = \int d\phi/\omega$	Axiom A4/A5
Length	ℓ_{ij}	Axiom A3
Direction	θ_{ij}	Axiom A4
Phase	ϕ_{ij}	Axiom A4
Interaction Strength	κ_{ij}	Axiom A5
Stress	$s_{ij} = \kappa_{ij}\ell_{ij}$	Axiom A5

Topological Invariants	β_n, χ	Axiom A6
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6 结论

6 Conclusion

中文

本文提出了宇宙的角度动量链接网络模型，并建立了完整的 ANG 公理体系。该模型的核心命题是：

- 最底层节点质量为零** —— 质量是闭合链接网络的涌现属性
- 链接杆携带能量、长度、角度、相位** —— 一切物理量来源于链接的四种属性
- 质量 = 闭合体内部链接网络** —— 质量是链接网络的总能量
- 力 = 闭合体界面链接分布** —— 力是边界链接的应力能
- 能量 = 闭合体运动 = 链接重连事件** —— 运动是重连事件的累积
- 时间 = 相位演化速率** —— 时间是链接相位的演化率

ANG 公理体系由六条公理（A1-A6）构成，统一了：

- 图论（网络结构）
- 度量几何（长度与角度）
- 连续介质力学（刚度与应力）
- 拓扑学（不变量）
- 非平衡动力学（重连演化）

该体系自然退化到广义相对论、量子场论和经典力学，为物理学的统一提供了一种基于网络几何的本体论基础。

English

This paper proposes an angular momentum link network model of the universe and establishes the complete ANG axiomatic system. The core propositions of the model are listed below:

- Fundamental nodes carry zero mass** — mass emerges as a collective property of closed link networks
- Link rods carry energy, length, angle and phase** — all physical quantities originate from these four link attributes

3. **Mass = internal link network within closed manifold** — mass equals the total energy contained in internal links
4. **Force = distribution of links on manifold boundary** — force arises from stress energy of boundary links
5. **Energy = motion of closed manifolds = link reconnection events** — motion corresponds to accumulated reconnection processes
6. **Time = phase evolution rate** — time is defined as the evolving phase of link structures

The ANG axiomatic system consists of six axioms (A1–A6), unifying:

- Graph theory (network structure)
- Metric geometry (length and angular relations)
- Continuum mechanics (stiffness and stress)
- Topology (topological invariants)
- Non-equilibrium dynamics (reconnection evolution)

This framework naturally reduces to general relativity, quantum field theory and classical mechanics, offering a network-geometry-based ontological foundation for the unification of physics.

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  — 宇宙不是由粒子构成的，而是由角动量链接杆组成的网络。质量是闭合链接网络的涌现，力是界面链接的分布，能量是重连事件的累积，时间是相位演化的速

率。

□   — The universe is not built from particles, but from a network of angular momentum link rods. Mass emerges from closed link networks, force originates from boundary link distributions, energy accumulates via reconnection events, and time is the evolution rate of phase.